On the Polynomial $D^4 - 2D^3 + 2D + 1$ and its Relation to the Fine Structure Constant

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Abstract

We report that the polynomial $P(D) = D^4 - 2D^3 + 2D + 1$ evaluated at $D = 4$ equals exactly 137, differing from the measured fine structure constant inverse $\alpha^{-1} = 137.035999084$ by only 0.026%. We systematically evaluate $P(D)$ for integer $D \in [1, 10]$ and find: (1) $P(D)$ yields primes for even $D \in \{2, 4, 6, 8\}$, but this pattern **fails** at $D = 10$; (2) $P(1), P(2), P(3)$ are Fibonacci numbers; (3) within Standard Model QED, the running coupling $\alpha^{-1}(\mu)$ cannot reach exactly 137 at any energy scale. Critically, we show that 138 distinct polynomials with small integer coefficients yield $Q(4) = 137$, demonstrating **non-uniqueness**. We present these as mathematical observations and discuss the criteria required for physical significance.

1 Introduction

The fine structure constant $\alpha \approx 1/137$ has attracted attention since Sommerfeld's introduction of the concept in 1916. The proximity of α^{-1} to the integer 137 has inspired numerous attempts to derive this value from first principles, from Eddington's numerological efforts to modern approaches in string theory.

Any claimed "derivation" of α^{-1} must satisfy stringent criteria:

1. The formula must be uniquely determined, not one of infinitely many fits
2. It must make testable predictions beyond reproducing known values
3. It should arise from a physical mechanism, not post-hoc numerology

This paper presents a mathematical observation—that a specific polynomial yields 137 at $D = 4$—and critically assesses what would be required for this to have physical significance.

2 The Polynomial Identity

Consider the polynomial

$$P(D) = D^4 - 2D^3 + 2D + 1 \tag{1}$$

Direct computation at $D = 4$:

$$\begin{aligned} P(4) &= 4^4 - 2 \cdot 4^3 + 2 \cdot 4 + 1 \\ &= 256 - 128 + 8 + 1 \\ &= \boxed{137} \end{aligned} \tag{2}$$

This is an exact algebraic identity, verifiable by arithmetic.

2.1 Comparison with Measured Value

The current CODATA 2022 recommended value is:

$$\alpha^{-1} = 137.035\,999\,177(21) \tag{3}$$

The relative deviation from 137:

$$\frac{|\alpha^{-1} - 137|}{137} = \frac{0.036}{137} = 2.6 \times 10^{-4} \quad (0.026\%) \tag{4}$$

3 Systematic Evaluation of $P(D)$

We evaluate $P(D)$ for integer dimensions $D \in [1, 10]$:

D	Computation	$P(D)$	Properties
1	$1 - 2 + 2 + 1$	2	Prime, F_3
2	$16 - 16 + 4 + 1$	5	Prime, F_5
3	$81 - 54 + 6 + 1$	34	F_9
4	$256 - 128 + 8 + 1$	137	Prime
5	$625 - 250 + 10 + 1$	386	2×193
6	$1296 - 432 + 12 + 1$	877	Prime
7	$2401 - 686 + 14 + 1$	1730	$2 \times 5 \times 173$
8	$4096 - 1024 + 16 + 1$	3089	Prime
9	$6561 - 1458 + 18 + 1$	5122	$2 \times 13 \times 197$
10	$10000 - 2000 + 20 + 1$	8021	13×617
11	$14641 - 2662 + 22 + 1$	12002	$2 \times 17 \times 353$
12	$20736 - 3456 + 24 + 1$	17305	5×3461

Table 1: Values of $P(D) = D^4 - 2D^3 + 2D + 1$ for $D \in [1, 12]$. F_n denotes the n-th Fibonacci number.

3.1 Observations

Observation 1 (Primality at even D): For $D \in \{2, 4, 6, 8\}$, all values $P(D) \in \{5, 137, 877, 3089\}$ are prime.

Observation 2 (Fibonacci at small D): $P(1) = 2 = F_3$, $P(2) = 5 = F_5$, and $P(3) = 34 = F_9$ are Fibonacci numbers.

Observation 3 (Composite at odd $D > 3$): For $D \in \{5, 7\}$, $P(D)$ is composite.

3.2 Counterexamples: $D = 10, 12$

The primality pattern **fails** at $D = 10$ and $D = 12$:

$$P(10) = 10000 - 2000 + 20 + 1 = 8021 = 13 \times 617 \quad \text{(composite)} \tag{5}$$

$$P(12) = 20736 - 3456 + 24 + 1 = 17305 = 5 \times 3461 \quad \text{(composite)} \tag{6}$$

This demonstrates that "$P(D)$ prime for even D" is **not** a general theorem—the pattern fails at multiple even values.

4 QED Running and the Integer Crossing Hypothesis

A natural question arises: could $\alpha^{-1}(\mu)$ equal exactly 137 at some energy scale μ through renormalization group (RG) running?

4.1 One-Loop QED Running

The QED beta function at one loop:

$$\mu \frac{d\alpha}{d\mu} = \frac{2\alpha^2}{3\pi} \sum_f Q_f^2 \, \theta(\mu - m_f) \tag{7}$$

where the sum runs over charged fermions f with charge Q_f and mass m_f.
Integrating, the running coupling satisfies:

$$\alpha^{-1}(\mu_2) = \alpha^{-1}(\mu_1) - \frac{2}{3\pi} \sum_f Q_f^2 \ln \frac{\mu_2}{\mu_1} \tag{8}$$

for $\mu_1, \mu_2 > m_f$.

4.2 Running Direction

Key property: $\alpha^{-1}(\mu)$ **decreases** monotonically with increasing μ:

- At $\mu = m_e$: $\alpha^{-1}(m_e) \approx 137.036$
- At $\mu = M_Z$: $\alpha^{-1}(M_Z) \approx 127.9$

4.3 Behavior Below m_e

Below the electron mass threshold:

- The electron contribution decouples from RG running
- Running is strongly suppressed (up to scheme-dependent low-energy effects)
- α^{-1} remains essentially frozen at ≈ 137.036

4.4 Conclusion

Result: Within Standard Model QED, there exists no energy scale μ where $\alpha^{-1}(\mu) = 137$ exactly.

The measured value $\alpha^{-1}(m_e) = 137.036$ is already above 137, and running only decreases the value at higher energies. This rules out RG running as a mechanism to explain the proximity to 137.

5 Critical Assessment

5.1 What Has Been Established

1. **Mathematical identity**: $P(4) = 137$ (exact)
2. **Empirical proximity**: $\alpha^{-1} = 137.036$, error 0.026%
3. **Primality pattern**: $P(D)$ prime for even $D \in \{2, 4, 6, 8\}$ (tested range)
4. **Falsified hypothesis**: No RG crossing at 137 in SM QED

5.2 What Has NOT Been Established

1. Why spacetime dimension $D = 4$ should enter this formula
2. Why this particular polynomial (and not others)
3. Any predictive power beyond fitting known α^{-1}
4. Physical mechanism connecting $P(D)$ to electromagnetic coupling

5.3 Non-Uniqueness

Exhaustive enumeration depends on the coefficient model:

- **Monic quartic** ($Q(D) = D^4 + aD^3 + bD^2 + cD + d$, $a, b, c, d \in [-10, 10]$): **138** solutions
- **General quartic** ($Q(D) = \sum_{k=0}^{4} a_k D^k$, $a_k \in [-10, 10]$): **716** solutions

The polynomial $P(D) = D^4 - 2D^3 + 2D + 1$ is **not unique** in producing 137 at $D = 4$. Any physical interpretation must specify a principled mechanism selecting a particular coefficient model and polynomial.

5.4 Comparison with Other Approximations

Expression	Value	Error from α^{-1}
$D^4 - 2D^3 + 2D + 1$ at $D = 4$	137.000	0.026%
$\frac{2}{3}(m_\mu/m_e)$	137.85	0.59%
44π	138.23	0.87%
$12(m_p/m_e)^{1/3}$	146.94	7.2%

Table 2: Various expressions approximating α^{-1}.

The polynomial $P(4)$ provides the closest integer approximation, but this alone does not establish physical significance.

6 What Would Constitute Evidence

For the polynomial identity to have physical meaning, one would need:

1. **Derivation**: A theoretical framework that predicts $\alpha^{-1} = P(D)$ for spacetime dimension D
2. **Prediction**: The formula should predict something new and testable, e.g.:
   - A higher-order correction to α^{-1}
   - The value of another coupling constant
   - Behavior in effective theories with $D \neq 4$
3. **Uniqueness**: Demonstration that $P(D)$ is not one of many polynomials that happen to give ≈ 137 at $D = 4$

Without these, the observation remains a numerical coincidence—interesting but not evidence of underlying physics.

7 Summary

We have presented the following results:

Statement	**Status**
$P(4) = 4^4 - 2 \cdot 4^3 + 2 \cdot 4 + 1 = 137$	Proved (arithmetic)
$\lvert\alpha^{-1} - 137\rvert/137 = 0.026\%$	Measured (CODATA 2022)
$P(D)$ prime for even D	Observed for $D \in \{2, 4, 6, 8\}$; **fails** at $D = 10$
$P(1), P(2), P(3) \in \{F_n\}$	Observed (3 values)
$\exists \mu : \alpha^{-1}(\mu) = 137$ in SM	Falsified
$P(D)$ is unique	**False**: 138 polynomials give $Q(4) = 137$

The falsification of the RG crossing hypothesis narrows the space of explanations: if running cannot produce 137, any connection between α^{-1} and 137 must arise from a different mechanism.

Whether the polynomial $P(D) = D^4 - 2D^3 + 2D + 1$ has deeper significance, or is merely a numerical coincidence in the vast space of polynomials, remains an open question. We have presented the mathematical facts; physical interpretation awaits a theoretical framework that can derive, rather than assume, this relationship.